NNN-T-360B
December 8, 1976
SUPERSEDING
NNN-T-360A
March 30, 1967

FEDERAL SPECIFICATION

TISSUE GRINDER, TENBROECK

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a glass Tenbroeck tissue grinder suitable for grinding small masses of tissue, especially in virus work.

1.2 Classification.

1.2.1 Sizes. The glass Tenbroeck tissue grinder covered by this specification shall be of the following sizes (see 6.1):

Large
Small

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-G-541	-	Glass (Laboratory).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-76	-	Boxes, Setup.

FSC 6640

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Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Document, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta Chiacgo, Kansas City, MO, fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, Standards and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard:

C 162-Standard Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

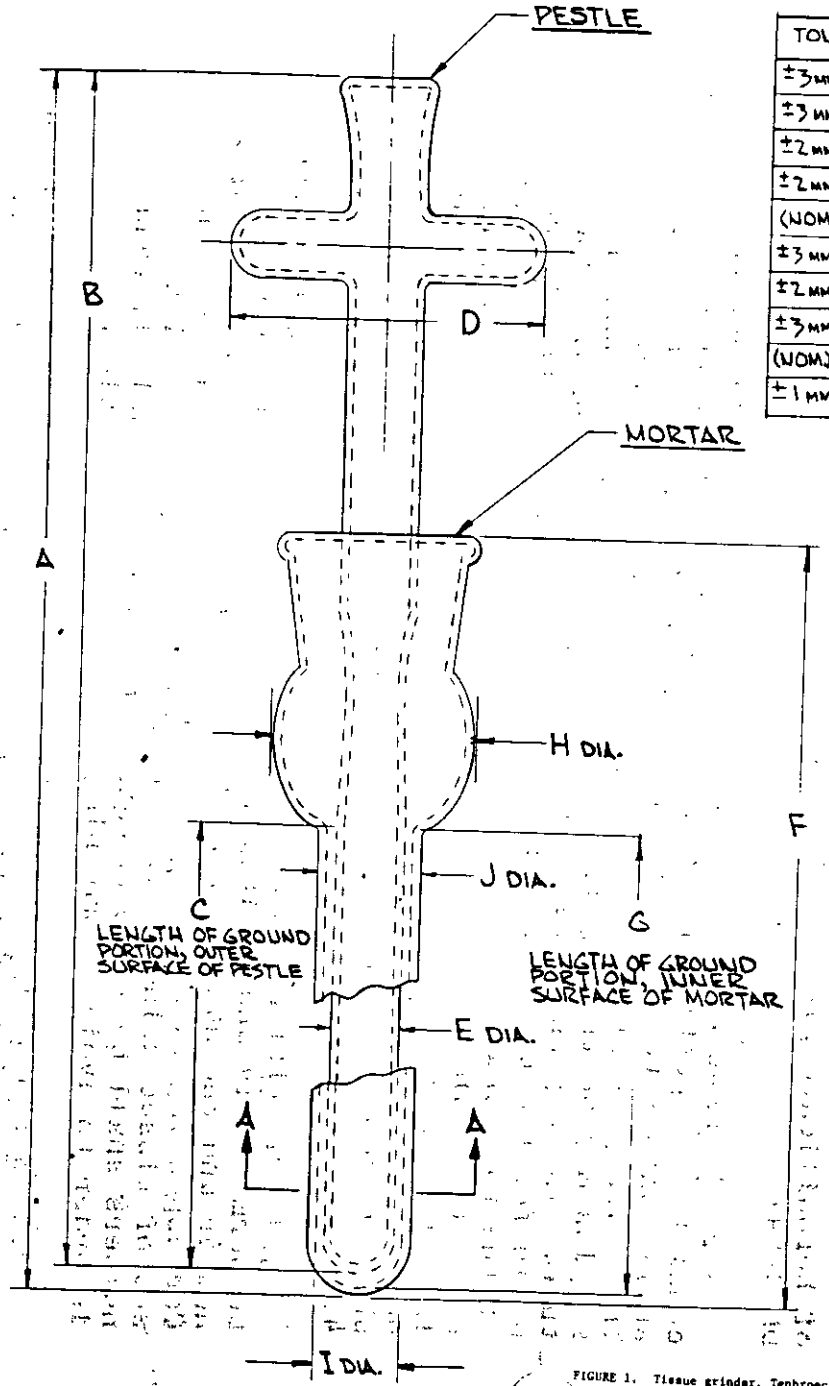
3.1 Material. The Tenbroeck tissue grinder shall be made of borosilicate glass conforming to the requirements of DD-G-541, type I, class A.

3.2 Style and design. The Tenbroeck tissue grinder shall consist of a pestle with handle and mortar. The mortar and pestle shall conform to figure 1 for style, design and dimensions. The lower part of the mortar shall be tubular so that tissue samples may be thoroughly washed between the pestle and the tube. The top shall be bowl-shaped to serve as an overflow basin for the tissue samples. The pestle and inside surface of the tube shall have a ground finish (see figure 1) similar to that obtained by grinding with an abrasive of 80-120 fineness. In addition, to insure proper maceration of tissues, the difference between the ground surface diameters of the mortar and pestle shall be 0.12 to 0.17 millimeters (mm.) when tested as specified in 4.4.1.

3.3 Identification markings. Each mortar and pestle shall be permanently and legibly marked with an identification number and the name or registered trademark of the manufacturer. Corresponding identification numbers shall be used for each set of tissue grinders (see 4.2.2). The identification numbers shall be large enough to be easily and quickly read in order to avoid mismatching of mortars and pestles.

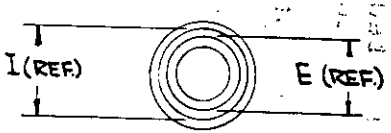
3.4 Workmanship. Workmanship shall be first class throughout. The tissue grinders shall be uniformly made and shall be free from blisters, bubbles, cracks, chips, seeds, striae, tears, checks, crizzles, stones and other defects that affect their serviceability. The tissue grinders shall be clean, dry and ready for use.

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TOL.	DIM.	LARGE	SMALL
±3 MM	A	340 MM	200 MM
±3 MM	B	335 MM	196 MM
±2 MM	C	130 MM	80 MM
±2 MM	D	90 MM	51 MM
(NOM)	E	20 MM	11 MM
±3 MM	F	210 MM	124 MM
±2 MM	G	130 MM	80 MM
±3 MM	H	99 MM	30 MM
(NOM)	I	20 MM	11 MM
±1 MM	J	25 MM	16 MM

NOTE:
DIMENSION "E" (O.D. OF PESTLE) & DIMENSION "I" (I.D. OF MORTAR) SEE SECT. A-A.



SECT. A-A

DIFFERENCE BETWEEN GROUND SURFACES "F" AND "I" DIAMETERS SHALL BE 0.12 TO 0.17 MM

FIGURE 1. Tissue grinder, Tenbroeck

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificate of quality, supplied by the manufacturer or supplier of the glass material, may be furnished in lieu of actual performance of such testing by the contractor. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the stock number, the Item Identification, the name of the material, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for examination purposes shall be the mortar and pestle.

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TABLE I. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for testing purposes shall be a set of tissue grinders (see 3.3). The acceptance number shall be zero.

TABLE II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Difference in diameters	3.2	4.4.1	S-1

4.3 Examination. The tissue grinders shall be examined for compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III. The definitions of glass defects shall be in accordance with ASTM Standard C-162. Examination shall not be restricted to the possible defects listed in the table.

TABLE III. Classification of defects

Categories	Defects
Major:	
101	Component of grinder not present or not properly formed.
102	Identification number not complete, correct, legible and permanent.
103	Grinder not free of cracks, chips and breaks.
104	Grinder not free of crizzles and checks.
105	Grinder not free of sharp edges and protrusions.
106	Grinder not symmetrical.
107	Edges at open end of mortar not cut at 90 degrees to the vertical axis.
108	Ground surfaces of mortar and pestle not free of smooth areas.
109	Glass of mortar and pestle not uniformly distributed.
110	Grinder not clean, dry, free of foreign material, and ready for use.

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TABLE III. Classification of defects (cont'd)

Categories	Defects
Minor:	
201	Grinder not free of blisters and bubbles.
202	Grinder not free of scratches, seeds, stones and tears.
203	Grinder not free of rough surfaces.
204	Manufacturer's identification not complete, correct, legible, and permanent.

4.3.2 Dimensional examination. The grinder shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

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4.3.3.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-566, PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-676 the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the applicable box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Difference in diameters. The difference between ground surface diameters of the mortar and pestle, as indicated in figure 1, shall be determined using a toolmaker's microscope or other instrument of equal accuracy.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each tissue grinder shall be thoroughly clean and the mortar and pestle shall be individually wrapped in tissue paper or other suitable material. The wrapped components shall be cushioned with suitable cushioning material within a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate package. Six unit packages containing tissue grinders of one size only shall be packaged in a box conforming to PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Tissue grinders shall be clean and individually packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

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5.2.1 Level A. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Each wood box shall be lined with a case liner conforming to MIL-L-10547. Closure and sealing of the case liner shall be as specified in the appendix of MIL-L-10547. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged tissue grinders shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Interior packages and exterior containers shall be marked in accordance with MIL-STD-129. Marking shall include the legend:

"GLASS - HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of specification.
(b) Specify size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Federal requirements.

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Preparing activity:
DSA-DM

Project No. 6640-0960

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	MATERIAL PROCURED UNDER A [ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

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